TASK 3: Secure Coding Review Report

Objective:

The objective of this task is to review a Python login application, identify security vulnerabilities, and implement security improvements by following secure coding practices.

Original Python Code:

Output of Original Code:

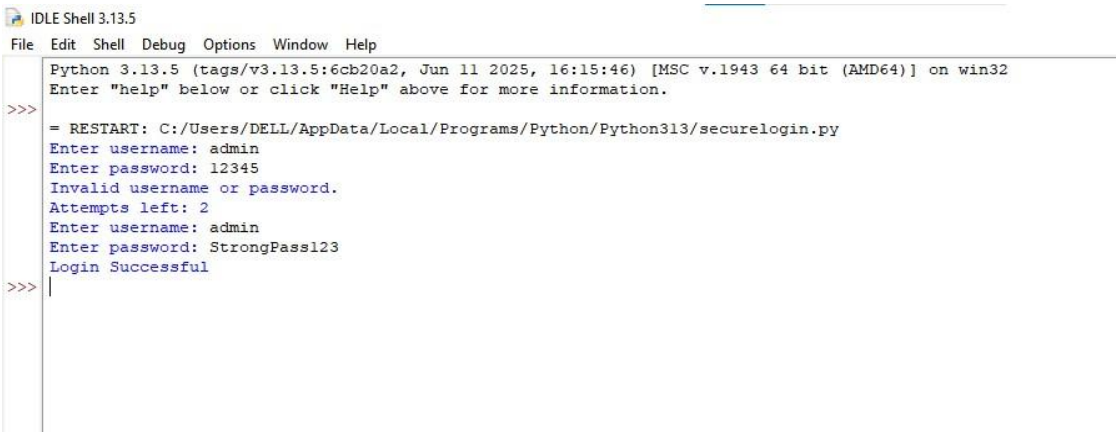
Security Vulnerabilities Identified:

During the code review, the following security issues were found:

- The username and password were hardcoded in the source code.
- The password was weak and easy to guess.
- Passwords were stored in plain text.
- There was no limit on login attempts, making the application vulnerable to brute-force attacks.

Modified Secure Python Code:

Output of Modified Code:



Improvements Made:

The following improvements were implemented:

- Replaced the weak password with a stronger password.
- Added a limit of three login attempts.
- Locked the account after three failed login attempts.
- Improved the overall security of the login process.

Recommendations:

- Use strong and unique passwords.
- Store passwords securely using hashing algorithms such as bcrypt or SHA-256.
- Avoid hardcoding credentials in source code.
- Store sensitive information in secure databases or configuration files.
- Implement account lockout and multi-factor authentication where possible.

Conclusion:

This secure coding review identified several common security vulnerabilities in the original Python login application. After implementing the recommended improvements, the application became more resistant to brute-force attacks and followed better security practices. This task helped improve understanding of secure coding principles and software security.